
EE/CprE/SE 4910 WEEKLY REPORT 1

9/16/2026 – 9/22/2026

Group number:

sdmay27-18

Project title: Eddy Current (EC) Probes for Conductivity Measurements

Client &/Advisor:

Advisor – Jiming Song

Client - Dr. Mingyang Lu, CNDE and ISU-ECpE

Team Members/Role:

Connor Crimmins - Client and advisor communication.

Jonathan Gatewood - Software and data-processing coordination.

Eric Williams - Circuit and measurement electronics coordination.

Truong LuuTeam organization - ,scheduling, and task tracking.

Sebastian De La Torr- Proband sensing hardware coordination.

Benjamin Ponce-Martinez - Testing, calibration, and documentation coordination.

For the project this week, our group got together to figure out how to approach this job before we talked to our advisor so that we had something to talk to him about. We did this by discussing previous projects of ours that were similar to what we worked on. So far, everyone has agreed with this plan. This week our goal is to talk to the advisor and begin designing/more research into our project. We plan to talk to the advisor about our plans and see what he thinks of the idea. We will look over components and software that we can use to design our circuit. We will also do some research on Eddy Currents so we can have a better understanding of the project.

- Eric Williams discussed projects he worked on for courses he took, such as EE330, where he built a temp sensor. He suggested that we create a circuit diagram for our device to start out and think through calculations for this project.
- Jonathan Gatewood added in his background working for Skyworks so that we can build and test our circuit until we have a perfect project. He discussed how we can use the lab equipment to do testing and validation work to optimize our circuit for the probe device.
- Benjamin Ponce-Martinez and Sebastian De La Torre discussed how we can acquire materials for the project. Both decided to buy different probes on Amazon to test with

until we have something that would work nicely.

- Truong Luu has been following up with everyone to make sure we all attend our weekly meetings and keep up to date with our project goals and deadlines.
- Connor Crimmins suggested previous experience working in research. Connor established a basis for our design that might be helpful for starting the project. He thought of creating an embedded circuit device that is essentially a measurement device that, through coding, we can accurately read and measure with the eddy current probes.

<u>NAME</u>	<u>Individual Contributions</u>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Member 1	Connor Crimmins	2	2
Member 2	Eric Williams	2	2
Member 3	Jonathan Gatewood	2	2
Member 4	Sebastian De La Torre	2	2
Member 5	Truong Luu	2	2
Member 6	Benjamin Ponce-Martinez	2.5	2.5

- Connor Crimmins will research the creation and assembling of a PCB that could be used with the eddy current probes to gather conductance information. Eric will also assist in this research.
- Eric Williams will research further into the utility of various probes and how to correlate them to a PCB that will be designed. Eric will update Connor on the findings to integrate the PCB design easier into use with the selected probes
- Jonathan Gatewood will research EC measurement principles, VNA operation, S-parameters, impedance, conductivity, and permeability. Sebastian will also assist in researching these topics.
- Sebastian De La Torre will collaborate with Jonathan on electromagnetic modeling and RF/impedance measurement methods; research commercial/custom coil geometries and high-frequency sensor frontends to establish test requirements for probe prototyping.
- Truong Luu will keep team members on track for their tasks and update all members on progress from each individual. He will also assist in any task where help is needed.
- Benjamin Ponce-Martinez will research coding to create software that can display information like conductance acquired from the hardware. Available team members will also assist in this process when they have time.